Trigonometric Equations

Concept

An inverse trig key gives one value. Use symmetry and periods to find all solutions.

General sine solutions

$x=a+2k\pi$ or $x=\pi-a+2k\pi$, k any integer.

Cosine / tangent

$x=\pm a+2k\pi$ for cosine; $x=a+k\pi$ for tangent.

Multiple arguments

Transform the interval for $bx+c$ before finding angles.

Quadratic trig

Solve in $\sin x$ or $\cos x$ first; reject outputs outside $[-1,1]$.

Example 1

Solve $\cos(2x) = 1/2$ on $0 \le x < 2\pi$.

Solution

1. The argument interval is $0 \le 2x < 4\pi$.
2. $2x = \pi/3, 5\pi/3, 7\pi/3, 11\pi/3$.
3. Divide by 2: $x = \pi/6, 5\pi/6, 7\pi/6, 11\pi/6$.

Trigonometric Equations

Example 2

Solve $2\cos^2 x - \cos x - 1 = 0$ on $0 \le x < 2\pi$.

Solution

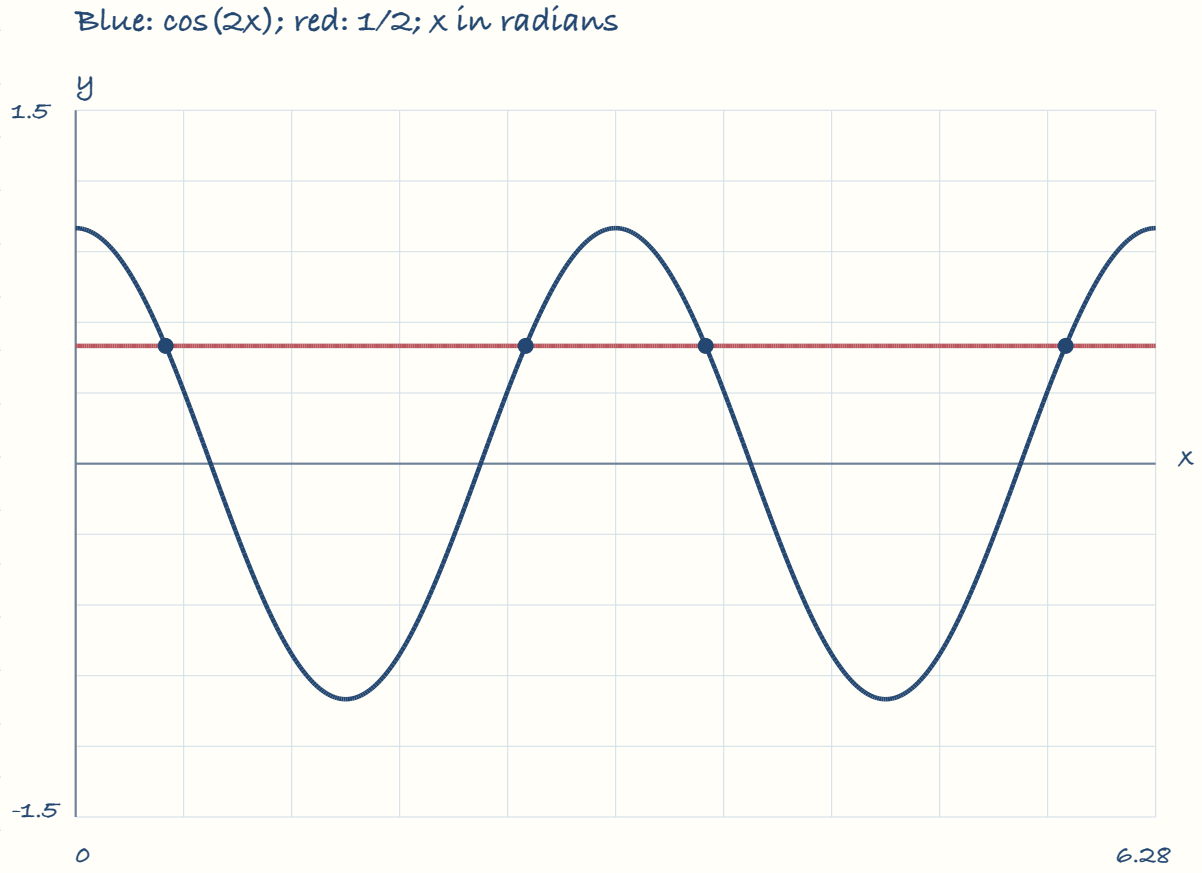
1. Factor: $(2\cos x + 1)(\cos x - 1) = 0$.

2. $\cos x = -1/2$ or 1.

3. $x = 2\pi/3, 4\pi/3, 0$. Do not include the excluded endpoint 2π.

Trigonometric Equations

Visual notes | read the graph



There are four intersections on $[0,2\pi)$. The doubled argument completes two cycles in this interval.

Try the picture

Locate intercepts and describe where the graph increases or decreases. Use the printed equation to verify at least two points. Explain any restrictions before extending the graph.

Trigonometric Equations

Practice | your turn

Show your steps. Include units and restrictions where needed.

1. Solve $\sin x = -\sqrt{2}/2$ on $[0,2\pi)$.

2. Solve $2\sin^2 x - 1 = 0$ on $[0,2\pi)$.

3. Solve $\tan(2x) = 0$ for all real x.

4. Does $\cos x = 1.2$ have a real solution?

5. Solve $\cos x = 0$ on $[0,2\pi]$.

Trigonometric Equations

Practice answers | check after trying

1. $5\pi/4$ and $7\pi/4$.

2. $\sin x = \pm\sqrt{2}/2$, so $\pi/4, 3\pi/4, 5\pi/4, 7\pi/4$.

3. $2x = k\pi$ gives $x = k\pi/2$, k any integer.

4. No: cosine is between -1 and 1.

5. $\pi/2$ and $3\pi/2$; neither endpoint solves the equation.